

An Quantum Algorithm for Computing General Pisano Periods, Wall-Sun-Sun Primes, and the Golden Mean

M. Chowdhury

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Abstract

We give a first quantum algorithm for the Pisano period modulo some integer m in this is achieved by our "Quantum Zig Zag Theorem" presented in this paper which uses a link between the (a, b) Fibonacci series and Pisano Periods. Hence we give a new quantum algorithm to compute Pisano periods. Our algorithm also requires the computation of the Golden mean constant as a classical-quantum subroutine so we give a first hybrid classical-quantum algorithm subroutine to compute the Golden mean, G (Catalan's Constant) and the beta function at some integer values and various other mathematical constants. Our classical-quantum subroutine uses Grover's algorithm in a new way i.e. it is based on a modified Grover algorithm. Consequently we give a new algorithms for Wall-Sun-Sun Primes and near Wall-Sun-Sun primes. While our algorithms may not be the most efficient for computing the Golden Mean etc. we think they are interesting as it is still a new algorithm quantum to solve the problem.

1 Introduction

The Fibonacci sequence is defined as $F_0 = 0$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$. The Pisano period is defined as the Fibonacci sequence modulo the positive integer m denoted by $\pi(m)$. The Pisano period has been studied since 1877 when Lagrange first notices the terms of the Fibonacci sequence repeat every 60 terms modulo 10. It is well known that a single Pisano period has 1, 2 or 4 zeroes. The number of zeros in a Pisano period is the order of m . The rank of m is the position of the first zero in a Pisano Period. Let $m = p$ where p is a prime number a prime p such that p^2 divides $\pi(p)$ is known as a WSS (Wall-Sun-Sun) prime. An infinite amount of WSS primes are conjectured to exist though none are known.

It is known [2], prior to the Wiles proof of Fermat's Last Theorem, that finding a WSS primes is equivalent to finding a counter example of the Fermat Last Theorem conjecture and thus they were searched for in the past to find such a counter example.

A near WSS prime is defined as

$$F_{p-(\frac{p}{5})} \equiv Ap \bmod p^2 \quad (200)$$

where

$$\begin{aligned} (\frac{p}{5}) &= 1 \text{ if } p \equiv 1, 4 \bmod 5 \\ &= -1 \text{ if } p \equiv 2, 3 \bmod 5 \end{aligned} \quad (201)$$

Our quantum algorithm to compute Pisano period is described in two parts. The first part is an algorithm (quantum subroutine one) to compute the golden mean constant to some precision (we have more than one choice to do this but we present a new algorithm to do this which is a hybrid quantum-classical algorithm). The second algorithm (quantum subroutine two) is a quantum period finding subroutine to find the Pisano -applying known quantum period finding algorithm will not work for finding the Pisano periods; it requires the Pisano Periods to be transformed using our "quantum Zig-Zag" theorem described below. Both the above quantum subroutines (one and two) are described below in the sections below.

1.1 Quantum Subroutine One: An Algorithm for the Golden Mean

As a subroutine we need the n th term for the FS (Fibonacci Series) or the (a,b) FS ((a,b) Fibonacci Series) which is a known natural generalisation of the FS. The golden mean is the constant

$$\varphi = \frac{1 + \sqrt{5}}{2} \quad (202)$$

If it is known the n th term for the FS is the nearest integer of a power of the golden mean multiplied by $\frac{1}{\sqrt{5}}$ in other words

$$F_n = \left\lfloor \frac{\varphi^n}{\sqrt{5}} \right\rfloor \text{ for } n \geq 0 \quad (203)$$

so we need to compute φ to some accuracy to compute the n term accurately because we will be using arbitrary bit arithmetic. The above is

derived from the Binet formula for the FS but we actually require instead of a and b to be two arbitrary integers, for our quantum algorithm, for reason that will be explained later, in a known natural generalised version which is known as the (a, b) Fibonacci Series (see for example [8])

$$F_n = F_{n-1} + F_{n-2} \text{ with } F_0 = a \text{ and } F_1 = b \quad (204)$$

again this can be solved by solving for the roots r_1 and r_2 of the Characteristic equation $r^2 - r - 1 = 0$, similar as is done to derive Binet's equation, so for the n th term of the generalised FS with a and b we have

$$F_{n,a,b} = \frac{b - a\Psi}{\sqrt{5}}(\varphi)^n + (a - \frac{b - a\Psi}{\sqrt{5}})(\Psi)^n \quad (205)$$

$$= \frac{b - a\Psi}{\sqrt{5}}(\frac{1 + \sqrt{5}}{2})^n + (a - \frac{b - a\Psi}{\sqrt{5}})(\frac{1 - \sqrt{5}}{2})^n \quad (206)$$

where $\varphi = \frac{1+\sqrt{5}}{2}$ and $\Psi = \frac{1-\sqrt{5}}{2}$ and where $F_{n,a,b}$ denotes this generalised FB and with $a=0, b=1$ we recover the original FS i.e, $F_n = F_{n,1,1}$. Note that because $\Psi = \frac{1}{2} - (\varphi - \frac{1}{2})$ we can compute 206 with knowledge of the golden mean only φ this means we only need to give one algorithm, our new hybrid classical-quantum, to compute an irrational number to some precision where this irrational number is φ . So we will now describe how to compute φ using a quantum algorithm using our new hybrid classical-quantum algorithm for numerical integration.

[To do here-describe how to compute the golden mean to some precision].

1.2 Quantum Subroutine Two: Finding the Pisano Period Using a New Transformation

The naive approach of applying the QFT directly to an elements of an arbitrary Pisano Period sequence does not work. The reason for this is because the map from F_n to $F_n \bmod m$, for some integer m , is not an 1-1 or injective map when considering the elements in one cycle for a Pisano period. The QFT to find a period works properly when the map is injective above. Now we can construct a related periodic function, in a way "building on top of the original Pisano Period", by using equation 206 and the following observations. Firstly observe that for some a, b r_1 are integer valued in equation 206 requires only integer arithmetic e.g. when $(a, b) = (1, 1)$ gives the Lucas sequence which can be computed with integer arithmetic. This means we can compute the n th term $F_{n,a,b}$ using only integer arithmetic for some (a, b) and certainly $F_{n,a,b}$ can be computed using all rational arithmetic and knowing the single irrational $\sqrt{5}$. We observe that the Pisano period of the function, i.e. the period

of the sequence $F_{n,a,a} \bmod m$ for some integer m , is either near (in some magnitude) to the Pisano period we are finding or is the same as the Pisano period in the FS (with a significant probability) when $a \neq m$ (since $F_{n,m,m} = 1$) we want to find for some a . For example the smallest odd Pisano period is 8 for $m = 3$ and the Pisano period for various a and b is as follows $F_{n,1,1} = 8, F_{n,4,4} = 8, F_{n,5,5} = 8, F_{n,3,3} = 1$. Another (random) example is the Pisano period is 60 for $m = 10$ and the Pisano period for various a and b is as follows $F_{n,1,1} = 60, F_{n,2,2} = 60, F_{n,4,4} = 3, F_{n,5,5} = 20, F_{n,6,6} = 60, F_{n,7,7} = 60$. But not that, with high probability, the elements in the Pisano sequence $F_{n,a,a} \bmod m$ is generally not the same as $F_{n,a',a'} \bmod m$ for some integer $a' \neq m$ where $a \neq a'$. Thus computing the period of the following function which is based on our Quantum "Zig-Zag" Theorem which gives a version of the equation 207b below

$$F_n'' = \sum_{\substack{K=1 \\ \gcd(m,K)=1}}^2 [((B^K)F_{n,K,K}) \oplus c_{K,n}) + ((B^3)F_{n+1,3,3}) \oplus c_{K,n})] \quad (207a)$$

207a this is a special case/truncated version of equation 208b below (without the $\gcd(m, K) = 1$ term in the sum)

$$\begin{aligned} F_n'' = & \sum_{K=1}^2 [((B^K)F_{n,K,K}) \oplus c_{K,n}) + ((B^{K+3})F_{n+1,3,3}) \oplus c_{K,n})] + \\ & B^3 \left(\sum_{K=1}^2 [((B^K)F_{n+1,K+2,K+2}) \oplus c_{K,n}) + ((B^3)F_{n+2,6,6}) \oplus c_{K,n})] \right) + \\ & B^6 \left(\sum_{K=1}^2 [((B^K)F_{n+3,K+4,K+4}) \oplus c_{K,n}) + ((B^3)F_{n+5,9,9}) \oplus c_{K,n})] \right) + \\ & \dots \\ & B^{3U} \left(\sum_{K=1}^2 [((B^K)F_{n+U,K+U+1,K+U+1}) \oplus c_{K,n}) + ((B^3)F_{n+U+1,3(U+1),3(U+1)}) \oplus c_{K,n})] \right) \end{aligned} \quad (208b)$$

for some integer U and some integer B where $B \geq m$ is sufficient. (We may take the B 's to be equal to $c_{K,n}$ in the above but for simplicity we do not take the B 's to be equal to $c_{K,n}$.)

Theorem 1 (Quantum "Zig Zag" Theorem) *The period of the function F_n'' (which is a transformation of the function F_n used to compute the Pisano Period classically) can be computed by using the Quantum Fourier Transform for a suitable value of U in equation 210 below.*

In addition to the above when $U = O(m)$ is polynomial in m , e.g. $U = \lfloor \frac{2}{6}(2 + 2 \lfloor \log_2(m) \rfloor) \rfloor$, we have that this period can be computed in polynomial time and polynomial space on a quantum computer. In addition each term Z_J added, as described below, each term increases the probability exponentially that the period will be measured correctly (in other words only a polynomial number of term U -e.g. $U = O'(m)$ in Z_U). Proof is sketched below (though strong numerical evidence is given). To summarise the theorem, F_n can be transformed in Polynomial time and Polynomial space, into another function F_n'' (we refer to as "Zig Zag function") from which the Pisano period that can be computed in polynomial time and polynomial space using a Quantum Fourier Transform.

Define

$$Z_{J,n} = B^{3J} \left(\sum_{K=1}^2 [((B^K)F_{n+J,K+J+1,K+J+1}) \oplus c_{K,n}) + ((B^3)F_{n+J+1,3(J+1),3(J+1)}) \oplus c_{K,n})] \right) \quad (209)$$

then 208

$$\begin{aligned} F_n'' = & \sum_{K=1}^2 [((B^K)F_{n,K,K}) \oplus c_{K,n}) + ((B^{K+3})F_{n+1,3,3}) \oplus c_{K,n})] + \\ & B^3 \left(\sum_{K=1}^2 [((B^K)F_{n+1,K+2,K+2}) \oplus c_{K,n}) + ((B^3)F_{n+2,6,6}) \oplus c_{K,n})] \right) + \\ & B^6 \left(\sum_{K=1}^2 [((B^K)F_{n+3,K+4,K+4}) \oplus c_{K,n}) + ((B^3)F_{n+5,9,9}) \oplus c_{K,n})] \right) + \\ & \dots \\ & B^{3U} \left(\sum_{K=1}^2 [((B^K)F_{n+U,K+U+1,K+U+1}) \oplus c_{K,n}) + ((B^3)F_{n+U+1,3(U+1),3(U+1)}) \oplus c_{K,n})] \right) \end{aligned} \quad (208)$$

can be rewritten as

$$F_n'' = Z_{0,n} + Z_{1,n} + Z_{2,n} + \dots + Z_{U,n} \quad (210)$$

(the $((B^3)F_{n+2U+1,2+2U,2+2U}) \oplus c_{K,n})$) can be thought of as a "single Zig-Zag" (or in other words "a single tooth") correction term (for example is $U = 0$ and this correction term is ignored then the quantum algorithm doesn't work so it is essential), that makes the function converge to the period of F_n'' so in other words it makes the success probability more likely regarding recovering the period. Equation 210 can be thought of as being composed of "multiple Zig Zags" or in other words "multiple teeth". Each term $Z_{J,n}$, for some J , it can be informally

thought of as "a single Zig-Zag" adding to the function that increases the success probability of the period being correctly recovered". Each "Zig-Zag" term increases the probability exponentially that the period will be recovered correctly in other words the success probability is proportional to $O''(1 - \frac{1}{2^U})$ for some polynomial function O'' and for a suitable B and some constants c_K . 207 can be considered as a transformation of 206 so that it makes it possible to find the Pisano period on a quantum computer due to it being transformed into 207; additionally we may need Shor's quantum factoring algorithm to find the Pisano Period. 208 is a specialisation of 207 where $O(m) = \lfloor \frac{2}{6}(2 + 2 \lfloor \log_2(m) \rfloor) \rfloor$ and $c_K = 0$ for all K .

For example with $m = 3$ we have $\lfloor \frac{2}{6}(2 + 2 \lfloor \log_2(m) \rfloor) \rfloor = 2$ hence $U = 2$ and so 210 becomes a single term

$$F_n'' = \sum_{K=1}^2 [((B^K)F_{n,K,K}) \oplus c_{K,n}) + ((B^{K+3})F_{n+1,3,3}) \oplus c_{K,n}] \\ + B^3 (\sum_{K=1}^2 [((B^K)F_{n+1,K+2,K+2}) \oplus c_{K,n}) + ((B^3)F_{n+2,6,6}) \oplus c_{K,n}])$$

if we set $c_{K,n} = 0$ for all K, n above the above simplifies to

$$F_n'' = \sum_{K=1}^2 [((B^K)F_{n,K,K}) + ((B^{K+3})F_{n+1,3,3})] \\ + B^3 (\sum_{K=1}^2 [((B^K)F_{n+1,K+2,K+2}) + ((B^3)F_{n+2,6,6})])$$

we have for a sample value of $n=1,2,\dots,20$ and $B=10$ the Pisano period modulum m where $m=3$ for the six (a,b) Fibonacci Series in the above is as follows (commas are omitted and spaces given between the endpoints of the periods for readability in the sequences below):

$F_{n,1,1}$ generates the sequence 01120221 01120221 01120221
 $F_{n,2,2}$ generates the sequence 02210112 02210112 02210112
 $F_{n+1,3,3}$ generates the sequence 00000000 00000000 00000000
 $F_{n+1,4,4}$ generates the sequence 01120221 01120221 01120221
 $F_{n+1,5,5}$ generates the sequence 02210112 02210112 02210112
 $F_{n+2,6,6}$ generates the sequence 00000000 00000000 00000000

therefore if we compute $|n\rangle > |F_n''\rangle$ in two quantum registers that are entangled we where $0 \leq n \leq 15$ have the following superposition in the second register

$$\begin{aligned} & \frac{1}{4}(|000120\rangle + |120120\rangle + |120210\rangle + |210000\rangle + |000210\rangle \\ & + |210210\rangle + |210120\rangle + |120000\rangle \\ & + |000120\rangle + |120120\rangle + |120210\rangle + |210000\rangle + |000210\rangle \\ & + |210210\rangle + |210120\rangle + |120000\rangle) \end{aligned}$$

which is a periodic function with period 8 and for suitable large n -this period can be recovered efficiently, for such as superposition like the above, by the using the quantum Fourier transform-and in this can this is the same as the Pisano period for $m = 3$.

A generalisation of equation 208 is as follows 211 (we give without proof)

$$\begin{aligned} F_n'' = & \sum_{K=1}^2 [((B^K)F_{n+\sigma(0),\sigma(K),\sigma(K)}) \oplus c_{K,n}) + ((B^{K+3})F_{n+\sigma(1),\sigma(3),\sigma(3)}) \oplus c_{K,n}] + (208) \\ & B^3 \left(\sum_{K=1}^2 [((B^K)F_{n+\sigma(1),K+\sigma(2),K+\sigma(2)}) \oplus c_{K,n}) + ((B^3)F_{n+\sigma(2),\sigma(6),\sigma(6)}) \oplus c_{K,n}] \right) + \\ & B^6 \left(\sum_{K=1}^2 [((B^K)F_{n+\sigma(3),K+\sigma(4),K+\sigma(4)}) \oplus c_{K,n}) + ((B^3)F_{n+\sigma(5),\sigma(9),\sigma(9)}) \oplus c_{K,n}] \right) + \\ & \dots \\ & B^{3U} \left(\sum_{K=1}^2 [((B^K)F_{n+\sigma(U),K+\sigma(U+1),K+\sigma(U+1)}) \oplus c_{K,n}) + \right. \\ & \left. ((B^3)F_{n+\sigma(U+1),\sigma((3(U+1)),\sigma((3(U+1))))} \oplus c_{K,n})] \right) \end{aligned} \quad (1)$$

where σ is a permutation (a map from integers to integers).

Then the quantum period finding algorithm is as follows.

Step 1. Pick a value for B . This is a classical step. Choose the value of m for the Pisano Period of the Fibonacci Series. The next step assumes the value of d is still large enough so the period if its more than 1 this can be detected. Choose the values of the constants $c_{K,n}$ in F_n'' .

Step 2. We have a quantum computer with the register $|x\rangle |y\rangle$. Apply the transformation

$$|n, o\rangle \rightarrow |n, \sum_n F_n''\rangle \quad (220)$$

$$\begin{aligned}
F_n'' = & \sum_{K=1}^2 [((B^K)F_{n+\sigma(0),\sigma(K),\sigma(K)}) \oplus c_{K,n}) + ((B^{K+3})F_{n+\sigma(1),\sigma(3),\sigma(3)}) \oplus c_{K,n}] + \quad (208) \\
& B^3 \left(\sum_{K=1}^2 [((B^K)F_{n+\sigma(1),K+\sigma(2),K+\sigma(2)}) \oplus c_{K,n}) + ((B^3)F_{n+\sigma(2),\sigma(6),\sigma(6)}) \oplus c_{K,n}] \right) + \\
& B^6 \left(\sum_{K=1}^2 [((B^K)F_{n+\sigma(3),K+\sigma(4),K+\sigma(4)}) \oplus c_{K,n}) + ((B^3)F_{n+\sigma(5),\sigma(9),\sigma(9)}) \oplus c_{K,n}] \right) + \\
& \dots \\
& B^{3U} \left(\sum_{K=1}^2 [((B^K)F_{n+\sigma(U),K+\sigma(U+1),K+\sigma(U+1)}) \oplus c_{K,n}) + \right. \\
& \left. ((B^3)F_{n+\sigma(U+1),\sigma((3(U+1)),\sigma((3(U+1))))} \oplus c_{K,n})] \right)
\end{aligned}$$

the number of states in superposition in the seconds register will be a power of 2.

Step 3. Now measure the second register to observe k . Assume c is rational and we picked n and b so a rational expansion is detectable. Then the new state is

$$\Psi >= \frac{1}{\sqrt{M}} \sum_{a \in A} |a, k >$$

where A is the set of integer such that $a < q$ and $A = \{u, u + p, u + 2p, u + (M - 1)p\}$. Where the period p is the Pisano Period of the Fibonacci sequence modulo m or p is a multiple of the Pisano Period of the Fibonacci sequence modulo m . Thus M is of size approximate $\frac{q}{r}$ which is much greater than 1. Hence

$$\Psi >= \frac{1}{\sqrt{M}} \sum_{w=0}^{M-1} |u + wp, k >$$

Step 4. Apply the QFT to the second register. This collapses the

entangled first register. Then

$$\begin{aligned}
|\Psi\rangle &= \frac{1}{\sqrt{M}} \sum_{w=0}^{M-1} |u + wp, k\rangle \text{ becomes} \\
|\Psi\rangle &= \frac{1}{\sqrt{qM}} \sum_{v=0}^{q-1} \sum_{w=0}^{M-1} |e^{(2\pi i(u+wp)*\frac{1}{q})}|v, k\rangle \\
&= \frac{1}{\sqrt{qM}} \sum_{v=0}^{q-1} e^{2\pi i v u / q} \sum_{w=0}^{M-1} |e^{(2\pi i(0+wp)*\frac{1}{q})}|v, k\rangle \\
&= \frac{1}{\sqrt{qM}} \sum_{v=0}^{q-1} e^{2\pi i v u / q} \left(\sum_{w=0}^{M-1} \Omega^d \right) |v, k\rangle
\end{aligned}$$

where $\Omega = e^{2\pi i w p / q}$ which is a root of unity.

Step 5. Measure the first register. Then register of we observer the state $|s\rangle$ with probability

$$pr(s) = \frac{1}{Mq} \left| \left(\sum_{w=0}^{M-1} \Omega^d \right) \right|^2$$

Noting $\Omega = e^{2\pi i v p / q}$ is $\frac{ve}{q}$ is not close to an integer then power of Ω cancel out such that the observation is quite unlikely. In this case $\left(\sum_{w=0}^{M-1} \Omega^d \right)$ is a finite geometric progression with a small sum. If $\frac{ve}{q}$ is near an integer then $pr(s) \approx \frac{M}{Nq} \approx \frac{1}{q}$ since $\Omega \approx 1$. Then the probability of observation is an integer such that

$$\frac{ve}{q} = L \text{ for some integer } L$$

Step 6. From the observed values can be found by an using an algorithm for continued fractions. The input to the algorithm if

$$\frac{\text{Observed Value}}{q} = \frac{v}{q}$$

computing the convergent of $\frac{v}{q}$ we have $\frac{a_i}{b_i}$. Then the values b_i . which are the denominators of the convergent, then the period p will be some integer multiple of b_i for some i . This can be found in polynomial time.

Step 7. This step requires classical computations only. If the p , in step 6 does not meet the criteria of a correct Pisano period (e.g. it is odd

and m is not 2) then we repeat the algorithm, a (polynomial) number of times, until we find a correct Pisano period. If the algorithm gives a multiple of the

Pisano period then we can factorise the candidate for the Pisano Period, p , using Shor's factoring algorithm to find the Pisano period. If m was chosen to be a prime in step 1 and p^2 divides $\pi(p)$ then p is a Wall-Sun-Sun prime. If $F_{p-(\frac{p}{5})} \equiv Ap \pmod{p^2}$ for some A then p is a near Wall Sun Sun prime.

End of algorithm

Note similar a quantum algorithm can be derived for finding the Pisano Periods of (a,b) Fibonacci numbers for which Pell numbers are an example (as an $(2,1)$ Fibonacci sequence). We sketch out the modifications to above quantum algorithm as follows. To find the Pisano period of an (a,b) Fibonacci numbers we would use a term analogous to 209 such as

$$Z_{J,n} = B^{3J} \left(\sum_{K=1}^2 [((B^K)F_{n+J,aK,bK}) \oplus c_{K,n}) + ((B^K)F_{n+J+1,aJ+1,bJ+1}) \oplus c_{K,n}) \right] \quad (209c)$$

and consequently we can derive an analogous version of equation 210 using 209c. We sketch an algorithm for finding the rank or number of zeroes in some Pisano period. Note we can find the number of zeroes in the Pisano period by searching for strings that begin with 00 on the state given by equation 220. Or instead we may use Grover's algorithm instead on the state 220 and searching for the zeroes in the second register. Or we may use Grover algorithm on the state $\sum \kappa |n\rangle : \left\lfloor \frac{\varphi^n}{\sqrt{5}} \right\rfloor \pmod{m} >$ for n a power of 2 and κ some amplitude to find the number of zeroes of some Pisano period or the rank of it. Also since the Fibonacci sequence can be classically computed in polynomial time using integer only matrix arithmetic (e.g. with a 2 by 2 matrix) this means that we can derive a variant of our quantum algorithm based on classical matrix arithmetic.

1.3 Conclusion

We presented a new quantum algorithm to compute the Pisano Period by linking it to the (a,b) Fibonacci Series and using our Quantum "Zig Zag" theorem. The complexity of our algorithm is expected to give polynomial speed up for computing the Pisano Period. We also described a quantum algorithm for finding the Pisano period of (a,b) Fibonacci numbers.

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